Below Zero and Uncounted: A Satellite Estimate of Women and Girls Living Through Freezing Winters in Gilgit-Baltistan and Chitral

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Introduction

Research on gendered harm in Pakistan's climate disasters concentrates on lowland summer floods; a review of 55 studies found none on winter displacement. Yet women in the northern mountains manage menstruation, pregnancy and sanitation through months of frozen water, closed roads and little privacy, in districts also exposed to glacial-lake floods. The review found no count of how many women and girls live in these conditions. This study uses open satellite data to estimate that number for Gilgit-Baltistan and Chitral, and to show how unusual the current winter is, to inform SDGs 5, 3, 13 and 10.

Methodology

Population came from WorldPop 2020 age and sex grids (100 m, limited to mapped buildings): girls aged 10–14 and women aged 15–49. Night land-surface temperature came from MODIS Aqua (1 km, 2017–2026); freezing nights were counted for November–March and weighted by where women and girls live. Snow cover came from MODIS Terra 8-day composites, comparing 1 October–8 November 2026 with the same dates in 2001–2025 for each district. Processing used Google Earth Engine; the code and a dashboard are open. The estimates describe exposure, not health outcomes.

Findings, Outcomes and Implications

About 479,000 women and girls aged 10–49 live in the two regions (Gilgit-Baltistan 319,000; Chitral 161,000). Nearly all (99.8%) live where January nights average below freezing: 72% where they average below −5 °C and 51% below −10 °C. They face about 117 freezing nights each winter. This winter began with snow cover 11.9 points above normal in Gilgit-Baltistan, the second highest in 26 years, with record highs in five districts. Winter preparedness must include menstrual supplies, safe and warm washing spaces, and season-specific data on women.

SDGs: 5, 3, 13, 10. Data: WorldPop, NASA MODIS, geoBoundaries, via Google Earth Engine. Code and dashboard available on request.